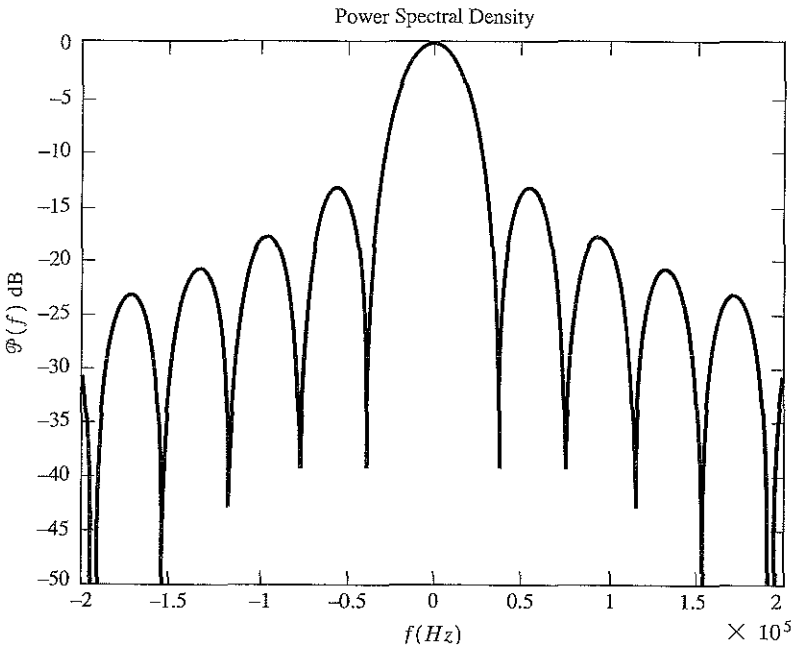


**Figure 3–48** PSD of an RS-232 signal with a data rate of 38,400 bits/s.

## PROBLEMS

**3–1** Demonstrate that the Fourier series coefficients for the switching waveform shown in Fig. 3–1b are given by Eq. (3–5b).

**3–2** **(a)** Sketch the naturally sampled PAM waveform that results from sampling a 1-kHz sine wave at a 4-kHz rate.

**(b)** Repeat part (a) for the case of a flat-topped PAM waveform.

★ **3–3** The spectrum of an analog audio signal is shown in Fig. P3–3. The waveform is to be sampled at a 10-kHz rate with a pulse width of $\tau = 50\ \mu s$.

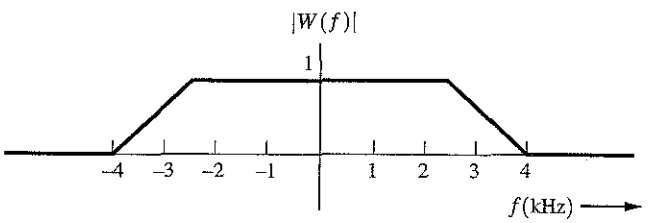


**Figure P3–3**

**(a)** Find an expression for the spectrum of the naturally sampled PAM waveform. Sketch your result.

**(b)** Find an expression for the spectrum of the flat-topped PAM waveform. Sketch your result.

**3–4** (a) Show that an analog output waveform (which is proportional to the original input analog waveform) may be recovered from a naturally sampled PAM waveform by using the demodulation technique shown in Fig. 3–4.

(b) Find the constant of proportionality, $C$, that is obtained with this demodulation technique, where $w(t)$ is the original waveform and $Cw(t)$ is the recovered waveform. Note that $C$ is a function of $n$, where the oscillator frequency is $nf_s$.

**3–5** Figure 3–4 illustrates how a naturally sampled PAM signal can be demodulated to recover the analog waveform by the use of a product detector. Show that the product detector can also be used to recover $w(t)$ from an instantaneously sampled PAM signal, provided that the appropriate filter $H(f)$ is used. Find the required $H(f)$ characteristic.

★ **3–6** Assume that an analog signal with the spectrum shown in Fig. P3–3 is to be transmitted over a PAM system that has ac coupling. Thus, a PAM pulse shape of the Manchester type, as given by Eq. (3–46a), is used. The PAM sampling frequency is 10 kHz. Find the spectrum for the Manchester encoded flat-topped PAM waveform. Sketch your result.

**3–7** In a binary PCM system, if the quantizing noise is not to exceed $\pm P$ percent of the peak-to-peak analog level, show that the number of bits in each PCM word needs to be

$$n \geq [\log_2 10]\left[\log_{10}\left(\frac{50}{P}\right)\right] = 3.32 \log_{10}\left(\frac{50}{P}\right)$$

(*Hint:* Look at Fig. 3–8c.)

★ **3–8** The information in an analog voltage waveform is to be transmitted over a PCM system with a $\pm 0.1\%$ accuracy (full scale). The analog waveform has an absolute bandwidth of 100 Hz and an amplitude range of $-10$ to $+10$ V.

(a) Determine the minimum sampling rate needed.

(b) Determine the number of bits needed in each PCM word.

(c) Determine the minimum bit rate required in the PCM signal.

(d) Determine the minimum absolute channel bandwidth required for transmission of this PCM signal.

**3–9** An 850-Mbyte hard disk is used to store PCM data. Suppose that a voice-frequency (VF) signal is sampled at 8 ksamples/s and the encoded PCM is to have an average SNR of at least 30 dB. How many minutes of VF conversation (i.e., PCM data) can be stored on the hard disk?

**3–10** An analog signal with a bandwidth of 4.2 MHz is to be converted into binary PCM and transmitted over a channel. The peak-signal quantizing noise ratio at the receiver output must be at least 55 dB.

(a) If we assume that $P_e = 0$ and that there is no ISI, what will be the word length and the number of quantizing steps needed?

(b) What will be the equivalent bit rate?

(c) What will be the channel null bandwidth required if rectangular pulse shapes are used?

★ **3–11** Compact disk (CD) players use 16-bit PCM, including one parity bit with 8 times oversampling of the analog signal. The analog signal bandwidth is 20 kHz.

(a) What is the null bandwidth of this PCM signal?

(b) Using Eq. (3–18), find the peak (SNR) in decibels.

**3–12** Given an audio signal with spectral components in the frequency band 300 to 3,000 Hz, assume that a sampling rate of 7 kHz will be used to generate a PCM signal. Design an appropriate PCM system as follows:

(a) Draw a block diagram of the PCM system, including the transmitter, channel, and receiver.
(b) Specify the number of uniform quantization steps needed and the channel null bandwidth required, assuming that the peak signal-to-noise ratio at the receiver output needs to be at least 30 dB and that polar NRZ signaling is used.
(c) Discuss how nonuniform quantization can be used to improve the performance of the system.

**3–13** The SNRs, as given by Eqs. (3–17a) and (3–17b), assume no ISI and no bit errors due to channel noise (i.e., $P_e = 0$). How large can $P_e$ become before Eqs. (3–17a) and (3–17b) are in error by 0.1% if $M = 4$, 8, or 16.

**★ 3–14** In a PCM system, the bit error rate due to channel noise is $10^{-4}$. Assume that the peak signal-to-noise ratio on the recovered analog signal needs to be at least 30 dB.
(a) Find the minimum number of quantizing steps that can be used to encode the analog signal into a PCM signal.
(b) If the original analog signal had an absolute bandwidth of 2.7 kHz, what is the null bandwidth of the PCM signal for the polar NRZ signaling case?

**3–15** Referring to Fig. 3–20 for a bit synchronizer using a square-law device, draw some typical waveforms that will appear in the bit synchronizer if a Manchester encoded PCM signal is present at the input. Discuss whether you would expect this bit synchronizer to work better for the Manchester encoded PCM signal or for a polar NRZ encoded PCM signal.

**3–16** (a) Sketch the complete $\mu = 10$ compressor characteristic that will handle input voltages over the range −5 to +5 V.
(b) Plot the corresponding expandor characteristic.
(c) Draw a 16-level nonuniform quantizer characteristic that corresponds to the $\mu = 10$ compression characteristic.

**3–17** For a 4-bit PCM system, calculate and sketch a plot of the output SNR (in decibels) as a function of the relative input level, $20 \log(x_{rms}/V)$ for
(a) A PCM system that uses $\mu = 10$ law companding.
(b) A PCM system that uses uniform quantization (no companding).
Which of these systems is better to use in practice? Why?

**★ 3–18** The performance of a $\mu = 255$ law companded PCM system is to be examined when the input consists of a sine wave having a peak value of $V$ volts. Assume that $M = 256$.
(a) Find an expression that describes the output SNR for this companded PCM system.
(b) Plot $(S/N)_{out}$ (in decibels) as a function of the relative input level, $20 \log(x_{rms}/V)$. Compare this result with that shown in Fig. 3–10.

**3–19** A multilevel digital communication system sends one of 16 possible levels over the channel every 0.8 ms.
(a) What is the number of bits corresponding to each level?
(b) What is the baud rate?
(c) What is the bit rate?

**3–20** A multilevel digital communication system is to operate at a data rate of 9,600 bits/s.
(a) If 4-bit words are encoded into each level for transmission over the channel, what is the minimum required bandwidth for the channel?
(b) Repeat part (a) for the case of 8-bit encoding into each level.

**★ 3–21** Consider a deterministic test pattern consisting of alternating binary 1's and binary 0's. Determine the magnitude spectra (not the PSD) for the following types of signaling formats as a function of $T_b$, the time needed to send one bit of data:

(a) Unipolar NRZ signaling.
(b) Unipolar RZ signaling where the pulse width is $\tau = \frac{3}{4}T_b$.
How would each of these magnitude spectra change if the test pattern was changed to an alternating sequence of four binary 1's followed by four binary 0's?

3–22 Consider a random data pattern consisting of binary 1's and 0's, where the probability of obtaining either a binary 1 or a binary 0 is $\frac{1}{2}$. Calculate the PSD for the following types of signaling formats as a function of $T_b$, the time needed to send one bit of data:
(a) Unipolar NRZ signaling.
(b) Unipolar RZ signaling where the pulse width is $\tau = \frac{3}{4}T_b$.
How do these PSDs for the random data cases compare to the magnitude spectra for the deterministic case of Prob. 3–21? What is the spectral efficiency for each of these cases?

3–23 Consider a deterministic data pattern consisting of alternating binary 1's and 0's. Determine the magnitude spectra (not the PSD) for the following types of signaling formats as a function of $T_b$, the time needed to send one bit of data:
(a) Polar NRZ signaling.
(b) Manchester NRZ signaling.
How would each of these magnitude spectra change if the test pattern was changed to an alternating sequence of four binary 1's followed by two binary 0's?

★ 3–24 Consider a random data pattern consisting of binary 1's and 0's, where the probability of obtaining either a binary 1 or a binary 0 is $\frac{1}{2}$. Calculate the PSD for the following types of signaling formats as a function of $T_b$, the time needed to send 1 bit of data:
(a) Polar RZ signaling where the pulse width is $\tau = \frac{1}{2}T_b$.
(b) Manchester RZ signaling where the pulse width is $\tau = \frac{1}{4}T_b$. What is the first null bandwidth of these signals? What is the spectral efficiency for each of these signaling cases?

3–25 Obtain the equations for the PSD of the bipolar NRZ and bipolar RZ (pulse width $\frac{1}{2}T_b$) line codes assuming peak values of ±3 V. Plot these PSD results for the case of $R = 1.544$ Mbits/s.

★ 3–26 In Fig. 3–16, the PSDs for several line codes are shown. These PSDs were derived assuming unity power for each signal so that the PSDs could be compared on an equal transmission power basis. Rederive the PSDs for these line codes, assuming that the peak level is unity (i.e., $A = 1$). Plot the PSDs so that the spectra can be compared on an equal peak-signal-level basis.

3–27 Using Eq. (3–36), determine the conditions required so that there are delta functions in the PSD for line codes. Discuss how this affects the design of bit synchronizers for these line codes. [*Hint:* Examine Eq. (3–43) and (6–70d).]

3–28 Consider a random data pattern consisting of binary 1's and 0's, where the probability of obtaining either a binary 1 or a binary 0 is $\frac{1}{2}$. Assume that these data are encoded into a polar-type waveform such that the pulse shape of each bit is given by

$$f(t) = \begin{cases} \cos\left(\dfrac{\pi t}{T_b}\right), & |t| < T_b/2 \\ 0, & |t| \text{ elsewhere} \end{cases}$$

where $T_b$ is the time needed to send one bit.
(a) Sketch a typical example of this waveform.
(b) Find the expression for the PSD of this waveform and sketch it.
(c) What is the spectral efficiency of this type of binary signal?

3–29 The data stream 01101000101 appears at the input of a differential encoder. Depending on the initial start-up condition of the encoder, find two possible differentially encoded data streams that can appear at the output.

**3–30** Create a practical block diagram for a differential encoding and decoding system. Explain how the system works by showing the encoding and decoding for the sequence 001111010001. Assume that the reference digit is a binary 1. Show that error propagation cannot occur.

**3–31** Design a regenerative repeater with its associated bit synchronizer for a polar RZ line code. Explain how your design works. (*Hint:* See Fig. 3–19 and the discussion of bit synchronizers.)

**3–32** Design a bit synchronizer for a Manchester NRZ line code by completing the following steps:
**(a)** Give a simplified block diagram.
**(b)** Explain how the synchronizer works.
**(c)** Specify the synchronizer's filter requirements.
**(d)** Explain the advantages and disadvantages of using this design for the Manchester NRZ line code compared with using a polar NRZ line code and its associated bit synchronizer.

**3–33** Figure 3–22c illustrates an eight-level multilevel signal. Assume that this line code is passed through a channel that filters the signal and adds some noise.
**(a)** Draw a picture of the eye pattern for the received waveform.
**(b)** Design a possible receiver with its associated symbol synchronizer for this line code.
**(c)** Explain how your receiver works.

★ **3–34** The information in an analog waveform is first encoded into binary PCM and then converted to a multilevel signal for transmission over the channel. The number of multilevels is eight. Assume that the analog signal has a bandwidth of 2,700 Hz and is to be reproduced at the receiver output with an accuracy of ±1% (full scale).
**(a)** Determine the minimum bit rate of the PCM signal.
**(b)** Determine the minimum baud rate of the multilevel signal.
**(c)** Determine the minimum absolute channel bandwidth required for transmission of this PCM signal.

★ **3–35** A binary waveform of 9,600 bits/s is converted into an octal (multilevel) waveform that is passed through a channel with a raised cosine-rolloff Nyquist filter characteristic. The channel has a conditioned (equalized) phase response out to 2.4 kHz.
**(a)** What is the baud rate of the multilevel signal?
**(b)** What is the rolloff factor of the filter characteristic?

**3–36** Assume that the spectral properties of an $L = 64$-level waveform with rectangular RZ-type pulse shapes are to be examined. The pulse shape is given by

$$f(t) = \Pi\left(\frac{2t}{T_s}\right)$$

where $T_s$ is the time needed to send one of the multilevel symbols.
**(a)** Determine the expression for the PSD for the case of equally likely levels where the peak signal levels for this multilevel waveform are + 10 V.
**(b)** What is the null bandwidth?
**(c)** What is the spectral efficiency?

**3–37** A binary communication system uses polar signaling. The overall impulse response is designed to be of the $(\sin x)/x$ type, as given by Eq. (3–67), so that there will be no ISI. The bit rate is $R = f_s = 300$ bits/s.
**(a)** What is the bandwidth of the polar signal?
**(b)** Plot the waveform of the polar signal at the system output when the input binary data is 01100101. Can you discern the data by looking at this polar waveform?

★ **3–38** Equation (3–67) gives one possible noncausal impulse response for a communication system that will have no ISI. For a causal approximation, select

$$h_e(t) = \frac{\sin \pi f_s(t - 4 \times 10^{-3})}{\pi f_s(t - 4 \times 10^{-3})} \Pi\left(\frac{t - 4 \times 10^{-3}}{8 \times 10^{-3}}\right)$$

where $f_s = 1{,}000$.

(a) Using a PC, calculate $H_e(f)$ by the use of the Fourier transform integral, and plot $|H_e(f)|$.

(b) What is the bandwidth of this causal approximation, and how does it compare with the bandwidth of the noncausal filter described by Eqs. (3–67) and (3–68)?

**3–39** Starting with Eq. (3–69), prove that the impulse response of the raised cosine rolloff filter is given by Eq. (3–73).

**3–40** Consider the raised cosine-rolloff Nyquist filter given by Eqs. (3–69) and (3–73).

(a) Plot $|H_e(f)|$ for the case of $r = 0.75$, indicating $f_1$, $f_0$, and $B$ on your sketch in a manner similar to Fig. 3–25.

(b) Plot $h_e(t)$ for the case of $r = 0.75$ in terms of $1/f_0$. Your plot should be similar to Fig. 3–26.

**3–41** Find the PSD of the waveform out of an $r = 0.5$ raised cosine-rolloff channel when the input is a polar NRZ signal. Assume that equally likely binary signaling is used and the channel bandwidth is just large enough to prevent ISI.

**★ 3–42** Equation (3–66) gives the condition for the absence of ISI (Nyquist's first method). Using that equation with $C = 1$ and $\tau = 0$, show that Nyquist's first method for eliminating ISI is also satisfied if

$$\sum_{k=-\infty}^{\infty} H_e\left(f + \frac{k}{T_s}\right) = T_s \qquad \text{for } |f| \le \frac{1}{2T_s}$$

**3–43** Using the results of Prob. 3–42, demonstrate that the following filter characteristics do or do not satisfy Nyquist's criterion for eliminating ISI ($f_s = 2f_0 = 2/T_0$).

(a) $H_e(f) = \dfrac{T_0}{2} \Pi\left(\dfrac{1}{2} fT_0\right)$.

(b) $H_e(f) = \dfrac{T_0}{2} \Pi\left(\dfrac{2}{3} fT_0\right)$.

**3–44** Assume that a pulse transmission system has the overall raised cosine-rolloff Nyquist filter characteristic described by Eq. (3–69).

(a) Find the $Y(f)$ Nyquist function of Eq. (3–75) corresponding to the raised cosine-rolloff Nyquist filter characteristic.

(b) Sketch $Y(f)$ for the case of $r = 0.75$.

(c) Sketch another $Y(f)$ that is not of the raised cosine-rolloff type, and determine the absolute bandwidth of the resulting Nyquist filter characteristic.

**★ 3–45** An analog signal is to be converted into a PCM signal that is a binary polar NRZ line code. The signal is transmitted over a channel that is absolutely bandlimited to 4 kHz. Assume that the PCM quantizer has 16 steps and that the overall equivalent system transfer function is of the raised cosine-rolloff type with $r = 0.5$.

(a) Find the maximum PCM bit rate that can be supported by this system without introducing ISI.

(b) Find the maximum bandwidth that can be permitted for the analog signal.

**3–46** Rework Prob. 3–45 for the case of a multilevel polar NRZ line code when the number of levels is four.

★ 3–47 Multilevel data with an equivalent bit rate of 2,400 bits/s is sent over a channel using a four-level line code that has a rectangular pulse shape at the output of the transmitter. The overall transmission system (i.e., the transmitter, channel, and receiver) has an $r = 0.5$ raised cosine-rolloff Nyquist filter characteristic.

(a) Find the baud rate of the received signal.

(b) Find the 6-dB bandwidth for this transmission system.

(c) Find the absolute bandwidth for the system.

3–48 Assume that a PCM-type system is to be designed such that an audio signal can be delivered at the receiver output. This audio signal is to have a bandwidth of 3,400 Hz and an SNR of at least 40 dB. Determine the bit rate requirements for a design that uses

(a) $\mu = 255$ companded PCM signaling.

(b) DPCM signaling.

Discuss which of the preceding systems would be used in your design and why.

3–49 Refer to Fig. 3–32, which shows typical DM waveforms. Draw an analog waveform that is different from the one shown in the figure. Draw the corresponding DM and integrator output waveforms. Denote the regions where slope overload noise dominates and where granular noise dominates.

3–50 A DM system is tested with a 10-kHz sinusoidal signal, 1 V peak to peak, at the input. The signal is sampled at 10 times the Nyquist rate.

(a) What is the step size required to prevent slope overload and to minimize granular noise?

(b) What is the PSD for the granular noise?

(c) If the receiver input is bandlimited to 200 kHz, what is the average signal–quantizing noise power ratio?

3–51 Assume that the input to a DM is $0.1t^8 - 5t + 2$. The step size of the DM is 1 V, and the sampler operates at 10 samples/s. Sketch the input waveform, the delta modulator output, and the integrator output over a time interval of 0 to 2 s. Denote the granular noise and slope overload regions.

3–52 Rework Prob. 3–51 for the case of an adaptive delta modulator where the step size is selected according to the number of successive binary 1's or 0's on the DM output. Assume that the step size is 1.5 V when there are four or more binary digits of the same sign, 1 V for the case of three successive digits, and 0.5 V for the case of two or fewer successive digits.

★ 3–53 A delta modulator is to be designed to transmit the information of an analog waveform that has a peak-to-peak level of 1 V and a bandwidth of 3.4 kHz. Assume that the waveform is to be transmitted over a channel whose frequency response is extremely poor above 1 MHz.

(a) Select the appropriate step size and sampling rate for a sine-wave test signal, and discuss the performance of the system, using the parameter values you have selected.

(b) If the DM system is to be used to transmit the information of a voice (analog) signal, select the appropriate step size when the sampling rate is 25 kHz. Discuss the performance of the system under these conditions.

3–54 One analog waveform $w_1(t)$ is bandlimited to 3 kHz, and another, $w_2(t)$, is bandlimited to 9 kHz. These two signals are to be sent by TDM over a PAM-type system.

(a) Determine the minimum sampling frequency for each signal, and design a TDM commutator and decommutator to accommodate the signals.

(b) Draw some typical waveforms for $w_1(t)$ and $w_2(t)$, and sketch the corresponding TDM PAM waveform.

★ 3–55 Three waveforms are time-division multiplexed over a channel using instantaneously sampled PAM. Assume that the PAM pulse width is very narrow and that each of the analog waveforms

is sampled every 0.15 s. Plot the (composite) TDM waveform when the input analog waveforms are

$$w_1(t) = 3\sin(2\pi t)$$

$$w_2(t) = \Pi\left(\frac{t-1}{2}\right)$$

and

$$w_3(t) = -\Lambda(t-1)$$

**3–56** Twenty-three analog signals, each with a bandwidth of 3.4 kHz, are sampled at an 8-kHz rate and multiplexed together with a synchronization channel (8 kHz) into a TDM PAM signal. This TDM signal is passed through a channel with an overall raised cosine-rolloff Nyquist filter characteristic of $r = 0.75$.

**(a)** Draw a block diagram for the system, indicating the $f_s$ of the commutator and the overall pulse rate of the TDM PAM signal.

**(b)** Evaluate the absolute bandwidth required for the channel.

**★ 3–57** Two flat-topped PAM signals are time-division multiplexed together to produce a composite TDM PAM signal that is transmitted over a channel. The first PAM signal is obtained from an analog signal that has a rectangular spectrum, $W_1(f) = \Pi(f/2B)$. The second PAM signal is obtained from an analog signal that has a triangular spectrum $W_2(f) = \Lambda(f/B)$, where $B = 3$ kHz.

**(a)** Determine the minimum sampling frequency for each signal, and design a TDM commutator and decommutator to accommodate these signals.

**(b)** Calculate and sketch the magnitude spectrum for the composite TDM PAM signal.

**3–58** Rework Prob. 3–56 for a TDM pulse code modulation system in which an 8-bit quantizer is used to generate the PCM words for each of the analog inputs and an 8-bit synchronization word is used in the synchronization channel.

**3–59** Design a TDM PCM system that will accommodate four 300-bit/s (synchronous) digital inputs and one analog input that has a bandwidth of 500 Hz. Assume that the analog samples will be encoded into 4-bit PCM words. Draw a block diagram for your design, analogous to Fig. 3–39, indicating the data rates at the various points on the diagram. Explain how your design works.

**★ 3–60** Design a TDM system that will accommodate two 2,400-bit/s synchronous digital inputs and an analog input that has a bandwidth of 2,700 Hz. Assume that the analog input is sampled at 1.11111 times the Nyquist rate and converted into 4-bit PCM words. Draw a block diagram for your design, and indicate the data rate at various points on your diagram. Explain how your TDM scheme works.

**3–61** Find the number of the following devices that could be accommodated by a T1-type TDM line if 1% of the line capacity were reserved for synchronization purposes.

**(a)** 110-bit/s teleprinter terminals.

**(b)** 8,000-bit/s speech codecs.

**(c)** 9,600-bit/s computer output ports.

**(d)** 64-kbit/s PCM VF lines.

**(e)** 144-kbit/s ISDN terminals.

How would these numbers change if each of the sources was operational an average of 10% of the time?

**3–62** Assume that a sine wave is sampled at four times the Nyquist rate using instantaneous sampling.
**(a)** Sketch the corresponding PWM signal.
**(b)** Sketch the corresponding PPM signal.

**3–63** Discuss why a PPM system requires a synchronizing signal, whereas PAM and PWM can be detected without the need for a synchronizing signal.

★ **3–64** Compare the bandwidth required to send a message by using PPM and PCM signaling. Assume that the digital source sends out 8 bits/character, so that it can send 256 different messages (characters). Assume that the source rate is 10 characters/s. Use the dimensionality theorem, $N/T_0 = 2B$, to determine the minimum bandwidth $B$ required.
**(a)** Determine the minimum bandwidth required for a PCM signal that will encode the source information.
**(b)** Determine the minimum bandwidth required for a PPM signal that will encode the source information.